

# Assignment 5

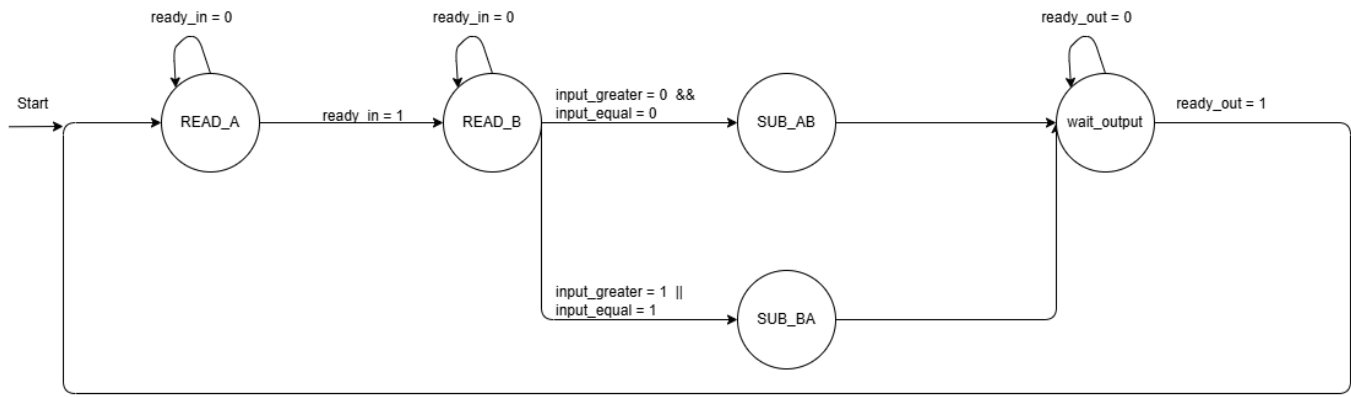
## Task 1

```
uint input(void)
{
    set_ready_in(1);
    while(!valid_in) wait();
    int val = read_bus_in();
    set_ready_in(0);
    return val;
}

void output(uint val)
{
    write_bus_out(val);
    set_valid_out(1);
    while(!ready_out) wait();
    set_valid_out(0);
}

void main(void)
{
    uint input_A, input_B, result;
    while(1)
    {
        input_A = input();
        input_B = input();
        result = (input_A > input_B) : input_A - input_B ? input_B - input_A;
        output(result);
    }
}
```

## Task 2



| state | next state | condition | actions |
|-------|------------|-----------|---------|
|-------|------------|-----------|---------|

| state       | next state  | condition                                        | actions                              |
|-------------|-------------|--------------------------------------------------|--------------------------------------|
| read_A      | read_A      | valid in = 0                                     | ready in <- 1                        |
| read_A      | read_B      | valid in = 1                                     | ready in <- 1 & write input to reg 0 |
| read_B      | read_B      | valid in = 0                                     | ready in <- 1                        |
| read_B      | subAB       | valid in = 1 & (input_greater or input_equal)    | ready in <- 1 & write input to reg 1 |
| read_B      | subBA       | valid in = 1 & not(input_greater or input_equal) | ready in <- 1 & write input to reg 1 |
| subAB       | wait_output | none                                             | Data_out = reg_0 - reg_1             |
| subBA       | wait_output | none                                             | Data_out = reg_1 - reg_0             |
| wait_output | wait_output | ready out = 0                                    | valid out = 1                        |
| wait_output | read_A      | ready out = 1                                    | valid out = 1                        |

### Task 3

```

main_statem_proc : process (state,valid_in,ready_out,input_equal,input_greater)
begin
    --default values
    --included at the as to make it unessesarry to specify in every state
    --this may hide errors, but prevents unintended latches
    opcode          <= alu_load;
    read_a_select    <= read_reg0;
    read_b_select    <= read_reg0;
    ready_in         <= '0';
    valid_out        <= '0';
    state_next       <= READ_A;
    write_select     <= write_none;

    --main implementation of statemachine
    case(state) is
        when READ_A =>
            ready_in <= '1';
            if(valid_in) then
                write_select <= write_reg0;
                read_a_select <= read_input;
                opcode <= alu_load;
                state_next <= READ_B;
            else
                state_next <= READ_A;
            end if;

        when READ_B =>
            ready_in <= '1';
            if(valid_in) then

```

```

        write_select <= write_reg1;
        read_a_select <= read_input;
        read_b_select <= read_reg0;
        opcode <= alu_load;

        if(input_greater or input_equal) then
            state_next <= SUB_AB;
        else
            state_next <= SUB_BA;
        end if;

    else
        state_next <= READ_B;
    end if;

when SUB_AB =>
    read_b_select <= read_reg0;
    read_a_select <= read_reg1;
    opcode <= alu_sub;
    write_select <= write_output;
    state_next <= WAIT_OUTPUT;

when SUB_BA =>
    read_b_select <= read_reg1;
    read_a_select <= read_reg0;
    opcode <= alu_sub;
    write_select <= write_output;
    state_next <= WAIT_OUTPUT;
when WAIT_OUTPUT =>
    valid_out <= '1';
    if(ready_out) then
        state_next <= READ_A;
    else
        state_next <= WAIT_OUTPUT;
    end if;
when others =>
    read_a_select <= read_reg0;
    read_b_select <= read_reg0;
    write_select <= write_none;
    valid_out <= '0';
    ready_in <= '0';
    opcode <= alu_load;
    state_next <= READ_A;
end case;
end process main_statem_proc;

```

## Task 4

The implemented design passed all tests.